

Rik L.E.M. Ubaghs

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Bio

Biomedical engineer and founder with expertise in neurotechnology, optical systems, medical devices, and data analytics. I specialize in developing and translating advanced imaging technologies into practical clinical applications, connecting cutting-edge research with meaningful medical impact.

Education

ETH Zurich, Zurich, Switzerland <i>Ph.D. Neurotechnology</i>	July 2017 - May 2023
ETH Zurich, Zurich, Switzerland <i>CAS Entrepreneurial Leadership in Technology Ventures</i>	Mar 2025 - Mar 2026
University of Amsterdam, Amsterdam, the Netherlands <i>Research Master Brain and Cognitive Sciences, Cum Laude</i>	Sept 2014 - June 2016
Maastricht University, Maastricht, the Netherlands <i>Bachelor Psychology and Neuroscience, Cum Laude</i>	Sept 2011 - June 2014

Skills

Languages:	Dutch (Native), English (Fluent), German (Basic)
Programming:	Python, Rust, Matlab, R
Databases/Querying:	SQL, MySQL
Skills:	Data Science, Biomedical Imaging, Optical Systems, Optical design (Zemax, Optiland) Mechanical Design (Autodesk, Solidworks)

Industry Positions

Arago Labs <i>Co-founder, CEO</i>	June 2023 - Current
• Founded an early-stage medical technology venture focused on improving surgical precision in oncology. • Designed advanced optical imaging platform combining nonlinear microscopy, and ML/DL-based tissue classification. • Defined market entry strategy, pricing model, reimbursement pathways, and go-to-market roadmap.	
Agora Technology, Maastricht, The Netherlands <i>Founder</i>	Nov 2016 - Feb 2018
• Built data solutions that aimed at streamlining/automating the workflow of medical facilities, especially focusing on surgical equipment. • We focused on internal logistics, and the creation, maintenance, and analysis of large-scale medical equipment databases using ML/DL.	

Research Positions

Inst. of Neuroinformatics, ETH Zurich, Zurich, Switzerland

July 2017 – June 2023

Ph.D. Candidate

- Development of a MRI compatible florescence microscope (full-cycle product design).
- Design of a paradigm to combine microscopy and MR imaging methods in awake rodents.
- Design of data pipelines for multi-modal data analysis.

Center for Neuroengineering, Duke University, Durham, USA

Dec 2015 – Nov 2016

Visiting Scientist

- Design of a hybrid brain decoding strategy in Non-human Primates.
- Development of ML/DL decoding pipeline for brain decoding.

Spinoza Center for Neuroimaging, Amsterdam, The Netherlands

Nov 2014 - Dec 2015

Junior Research Scientist

- MR operator.
- Design and supervision of several large-scale research projects.

Extracurricular Positions

Natural Movements, The Netherlands

March 2007 - May 2012

Co-Founder

- Natural Movements is a non-profit organization that aims to engage underprivileged youth in the local community through behavioral coaching and physical education.
- Worked together with social service organisations to provide youth with any additional support they required.

Selected publications and patents

Publications:

- **Ubaghs, R.L.E.M.**, ... Grewe, B.F. Simultaneous single-cell calcium imaging of neuronal population activity and brain-wide BOLD fMRI. *Nature Methods* (in press).
- Rychen, J., ... **Ubaghs, R.L.E.M.**, ... Hahnloser, R. Recordings of cetacean vocal behaviour with hydrophone arrays in Northern Norway. *bioRxiv* (2023); under review *Nature Scientific Data*. <https://whalesonic.stream>

Patents:

- Sainz Villalba, L., **Ubaghs, R.L.E.M.** DEVICE AND METHOD FOR PROVIDING HISTOLOGICAL NAVIGATION INFORMATION DURING IN-VIVO SURGERY. European Patent Application, Filed 2026.

Grants and fellowships

- ETH Zurich Pioneer Fellowship
- Swiss Cancer Foundation research grant
- Galbiati Foundation research donation

April 2025 - Current
September 2026
June 2025